

MOS Capacitor Tutorial Answer

1 (a) n-type

(b) inversion

2 (a) aluminium = -0.86 eV

n+ polysilicon = -0.91 eV

p+ polysilicon = $+0.21 \text{ eV}$

(c) n+ polysilicon = 0.272 V

p+ polysilicon = 1.39 V

3 (a) Yes, E_F is constant.

(c) $V_G = -0.877 \text{ V}$

(d) $C/C_{ox} = 0.503$

(f) $V_{TH} = -1.411 \text{ V}$

4. $N_A = 1.1 \times 10^{15} \text{ cm}^{-3}$